Monomial f(q) = a*qˆ7 Report

Polynomial

$$f(q) = aq^7$$

Naive

$$g_{\text{naive}}(X) = X^7 a$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 5040a\frac{\Delta}{\epsilon}^6 q + 840a\frac{\Delta}{\epsilon}^4 q^3 + 42a\frac{\Delta}{\epsilon}^2 q^5 + aq^7$$

$$\begin{aligned}\mathrm{Var}(g_{\text{naive}}(X)) = {} & 87178291200a^2\frac{\Delta}{\epsilon}^{14} + 43563744000a^2\frac{\Delta}{\epsilon}^{12} q^2 \\ & + 3623961600a^2\frac{\Delta}{\epsilon}^{10} q^4 + 119952000a^2\frac{\Delta}{\epsilon}^{8} q^6 \\ & + 2081520a^2\frac{\Delta}{\epsilon}^{6} q^8 + 20580a^2\frac{\Delta}{\epsilon}^{4} q^{10} + 98a^2\frac{\Delta}{\epsilon}^{2} q^{12}\end{aligned}$$

$$\begin{aligned}\mathrm{MSE}(g_{\text{naive}}) = {} & 87178291200a^2\frac{\Delta}{\epsilon}^{14} + 43589145600a^2\frac{\Delta}{\epsilon}^{12} q^2 \\ & + 3632428800a^2\frac{\Delta}{\epsilon}^{10} q^4 + 121080960a^2\frac{\Delta}{\epsilon}^{8} q^6 \\ & + 2152080a^2\frac{\Delta}{\epsilon}^{6} q^8 + 22344a^2\frac{\Delta}{\epsilon}^{4} q^{10} + 98a^2\frac{\Delta}{\epsilon}^{2} q^{12}\end{aligned}$$

$$\mathrm{Bias}(g_{\text{naive}}) = 5040a\frac{\Delta}{\epsilon}^6 q + 840a\frac{\Delta}{\epsilon}^4 q^3 + 42a\frac{\Delta}{\epsilon}^2 q^5$$

Unbiased

$$g_{\text{unb}}(X) = X^7 a - 42X^5 a\frac{\Delta}{\epsilon}^2$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^7$$

$$\begin{aligned}\mathrm{Var}(g_{\text{unb}}(X)) = {} & 53343360000a^2\frac{\Delta}{\epsilon}^{14} + 26671680000a^2\frac{\Delta}{\epsilon}^{12} q^2 \\ & + 2222640000a^2\frac{\Delta}{\epsilon}^{10} q^4 + 74088000a^2\frac{\Delta}{\epsilon}^{8} q^6 \\ & + 1323000a^2\frac{\Delta}{\epsilon}^{6} q^8 + 14700a^2\frac{\Delta}{\epsilon}^{4} q^{10} + 98a^2\frac{\Delta}{\epsilon}^{2} q^{12}\end{aligned}$$

$$
\begin{aligned}
\mathrm{MSE}(g_{\mathrm{unb}}) = {} & 53343360000a^2\frac{\Delta}{\epsilon}^{14} + 26671680000a^2\frac{\Delta}{\epsilon}^{12}q^2 \\
& + 2222640000a^2\frac{\Delta}{\epsilon}^{10}q^4 + 74088000a^2\frac{\Delta}{\epsilon}^{8}q^6 \\
& + 1323000a^2\frac{\Delta}{\epsilon}^{6}q^8 + 14700a^2\frac{\Delta}{\epsilon}^{4}q^{10} + 98a^2\frac{\Delta}{\epsilon}^{2}q^{12}
\end{aligned}
$$

Comparison

$$
\text{mean gap} = -5040a\frac{\Delta}{\epsilon}^{6}q - 840a\frac{\Delta}{\epsilon}^{4}q^3 - 42a\frac{\Delta}{\epsilon}^{2}q^5
$$

$$
\begin{aligned}
\text{variance gap} = {} & -33834931200a^2\frac{\Delta}{\epsilon}^{14} - 16892064000a^2\frac{\Delta}{\epsilon}^{12}q^2 \\
& - 1401321600a^2\frac{\Delta}{\epsilon}^{10}q^4 - 45864000a^2\frac{\Delta}{\epsilon}^{8}q^6 \\
& - 758520a^2\frac{\Delta}{\epsilon}^{6}q^8 - 5880a^2\frac{\Delta}{\epsilon}^{4}q^{10}
\end{aligned}
$$

$$
\begin{aligned}
\text{MSE gap} = {} & -33834931200a^2\frac{\Delta}{\epsilon}^{14} - 16917465600a^2\frac{\Delta}{\epsilon}^{12}q^2 - 1409788800a^2\frac{\Delta}{\epsilon}^{10}q^4 \\
& - 46992960a^2\frac{\Delta}{\epsilon}^{8}q^6 - 829080a^2\frac{\Delta}{\epsilon}^{6}q^8 - 7644a^2\frac{\Delta}{\epsilon}^{4}q^{10}
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{Bias}(g_{\mathrm{naive}})^2 = {} & 25401600a^2\frac{\Delta}{\epsilon}^{12}q^2 + 8467200a^2\frac{\Delta}{\epsilon}^{10}q^4 \\
& + 1128960a^2\frac{\Delta}{\epsilon}^{8}q^6 + 70560a^2\frac{\Delta}{\epsilon}^{6}q^8 + 1764a^2\frac{\Delta}{\epsilon}^{4}q^{10}
\end{aligned}
$$